Written HW3

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I use Neovim (BTW)

Full HW repository at
github.com/LoganCTanner/School

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1. Suppose the number of times a person contracts a cold in a given year is a Poisson with $\lambda = 5$. Also Suppose a new wonder drug has just been marketed that reduces the Poisson parameter to $\lambda = 3$ for 75% of the population; for the other 25%, the drug has no effect. If an individual tries the drug for a year, and gets 2 colds during that time, what is the probability that the drug is beneficial for this individual?

   Answer:
   To find the probability that the drug is beneficial for this individual we may use Bayes' Theorem to find the likelihood that that they are in the group of individuals where the poisson distribution is given by $\lambda = 3$.
   $P(\lambda = 3|\text{drug}) = .75$, $P(\lambda = 5|\text{drug}) = .25$
   $P(\lambda = 3|k = 2) = \frac{P(\lambda=3)P(k=2|\lambda=3)}{P(k=2)}$

$$P(k = 2) = P(\lambda = 3)P(k = 2|\lambda = 3) + P(\lambda = 5)P(k = 2|\lambda = 5)$$

$$= .75\left(e^{-3}\frac{3^2}{2!}\right) + .25\left(e^{-5}\frac{5^2}{2!}\right)$$

$$\approx .75 \times .224042 + .25 \times .0842243$$

$$\approx .189088$$

$$P(\lambda = 3|k = 2) \approx \frac{.75 \times .224042}{.189088}$$

$$\approx \boxed{.888641}$$

2. An Interviewer is given a list of people she can interview. She will conduct at most 5 interviews, and each person independently agrees to be interviewed with probability $\frac{2}{3}$.

   (a) If the list has 5 people on it, what's the probability that she gets five interviews?
   Answer:
   This question follows the format of a Binomial Distribution Probability for the simplest case where $k = n$
   $X \sim Binomial(5, \frac{2}{3})$
   $P(X = 5) = \binom{5}{5} \times \left(\frac{2}{3}\right)^5 \times \left(\frac{1}{3}\right)^0$

   $$= \boxed{\frac{32}{243}}$$

   (b) What if the list has 8 people on it?
   Answer:
   The question remains to be a binomial, though now with $0 < k < n$, resulting in
   $X \sim Binomial(8, \frac{2}{3})$
   $P(X = 5) = \binom{8}{5} \times \left(\frac{2}{3}\right)^5 \times \left(\frac{1}{3}\right)^3$

   $$= \boxed{\frac{1792}{6561}}$$

(c) If the list has 8 people on it, and she begins at the top, what is the probability that she interviews the 7th person?
Answer:
In this case there are 6 possible people preceding the 7th interviewee
$X \sim Binomial(6, \frac{2}{3})$

$$P(\text{7th person interviewed}) = \frac{2}{3} \times \Sigma_{i=0}^{4} P(X = i)$$

$$P(X = i) = \binom{6}{i} \times \left(\frac{2}{3}\right)^{i} \times \left(\frac{1}{3}\right)^{6-i}$$

$$\Sigma_{i=0}^{4} P(X = i) =$$

$$\binom{6}{0} \times \left(\frac{2}{3}\right)^{0} \times \left(\frac{1}{3}\right)^{6}$$

$$+\binom{6}{1} \times \left(\frac{2}{3}\right)^{1} \times \left(\frac{1}{3}\right)^{5}$$

$$+\binom{6}{2} \times \left(\frac{2}{3}\right)^{2} \times \left(\frac{1}{3}\right)^{4}$$

$$+\binom{6}{3} \times \left(\frac{2}{3}\right)^{3} \times \left(\frac{1}{3}\right)^{3}$$

$$+\binom{6}{4} \times \left(\frac{2}{3}\right)^{4} \times \left(\frac{1}{3}\right)^{2}$$

$$= \frac{1}{3^6} + \frac{6 \times 2}{3^6} + \frac{15 \times 2^2}{3^6} + \frac{20 \times 2^3}{3^6} + \frac{15 \times 2^4}{3^6}$$

$$= \frac{473}{729}$$

$$P(\text{7th person interviewed}) = \frac{2}{3} \times \frac{473}{729} = \boxed{\frac{946}{2187}} \approx 0.432556$$

To verify, I ran the following Python script, which returned 0.433336

```python
import random

count = 0
trials = 1000000

for _ in range(trials):
    interviews = 0

    # Go through persons 1-6
    for person in range(1, 7):
        if random.random() < 2/3:  # Person says yes
            interviews += 1
            if interviews == 5:  # Hit the limit
                break

    # Check if we can reach person 7
    if interviews < 5:
        # Ask person 7
        if random.random() < 2/3:  # Person 7 says yes
            count += 1

probability = count / trials
print(f"P(interviews person 7) = {probability}")

```

(d) As in part (c), what is the probability she interviews the 8th person?
Answer:
Like in (c), we are looking for the probability where the j^{th} positioned potential interviewee is interviewed, where now $j = 8$. We can then substitute in $j - 1 = 7$ for the number of preceding potential interviewees and recalculate to find:
$X \sim Binomial(7, \frac{2}{3})$
$P(\text{8th person interviewed}) = \frac{2}{3} \times \Sigma_{i=0}^{4} P(X = i)$
$P(X = i) = \binom{7}{i} \times \left(\frac{2}{3}\right)^{i} \times \left(\frac{1}{3}\right)^{7-i}$
$\Rightarrow P(\text{8th person interviewed}) = \boxed{\frac{1878}{6561}} \approx 0.28624$
Then, modifying the previous Python script to be range(1,8) on line 10 and recomputing, the approximation verifies with a result of 0.285745

3. A restaurant offers kids' meals that come with a toy. There are 5 different types of toy available, and each is equally likely to come with a given meal. Find the expected number of meals needed to collect all five types of toy. (Note that it may take more than 5 meals because getting a toy one already has is a possibility.)

   Answer:

   This may be considered as a sum of expected Binomial Distribution Probabilities, where the probability p is given by $p = \frac{n-k}{n}$ where $n = \#$ total toys possible $= 5$ and $k = \#$ total toys owned. Given that the Expected Value of a Probability of a Binomial Random Variable is calculated by $E[X] = np$, the sum of expectations can then be obtained by:

$$E[X] = \sum_{k=0}^{5} 5 \times \frac{5-k}{5} = \sum_{k=0}^{5} 5 - k = \boxed{15}$$

4. Again consider the restaurant above. Find the expected value and standard deviation of the number of types of toy still uncollected after 7 meals have been purchased.
Answer:
Given an indicator random variable Y_i that is 1 if the i^{th} type of toy is still uncollected after 7 purchases, the probability p of Y_i is $\frac{4}{5}^7$. Since the indicator random variable is a Bernoulli r. v., $E[Y_i] = p = \frac{4}{5}^7$. Then, the expected value for all Y_i becomes

$$E[Y] = E[Y_1 + \cdots + Y_5] = 5 \times \left(\frac{4}{5}\right)^7 = \boxed{\frac{4^7}{5^6} \approx 1.05}$$

Then to calculate std. dev. the variance can be obtained through

$$\mathrm{Var}(Y) = E[Y^2] - (E[Y])^2$$

$$\begin{aligned}
E[Y^2] &= E[Y_1 + \cdots + Y_5 + Y_1Y_2 + \cdots + Y_4Y_5] \\
&= 5\left(\frac{4}{5}\right)^7 + \binom{5}{2}\left(\frac{3}{5}\right)^7 \\
&= \frac{4^7}{5^6} + \frac{2 \times 3^7}{5^6} \\
(E[Y])^2 &= \frac{4^{14}}{5^{12}} \\
\mathrm{Var}(Y) &= \frac{4^7}{5^6} + \frac{2 \times 3^7}{5^6} - \frac{4^{14}}{5^{12}} \\
&\approx \boxed{0.229}
\end{aligned}$$

with the std. dev. then being

$$\sigma \approx \boxed{\sqrt{0.229} \approx 0.478}$$

5. Suppose that the lifetime of a certain type of electronic device is modeled using a continuous random variable, X whose probability density function is given by

$$f(x) = \begin{cases} A/100 & 0 < x < 10, \\ A/x^2 & x > 10 \end{cases}$$

with x given in years.

(a) Find A.
Answer:
Integrating over any probability density function yields 1, finding A then becomes a matter of integration and algebra

$$1 = \int_0^{10} A/100dx + \int_{10}^{\infty} A/x^2dx$$

$$1 = A/100[x]_0^{10} + A(-1)[1/x]_{10}^{\infty}$$

$$1 = A/10 + A/10 = A/5$$

$$\boxed{A = 5}$$

(b) Find $PX > 20$. Given the cases provided, and $20 > 10$, the solution becomes an elementary integral

$$P\{X > 20\} = \int_{20}^{\infty} \frac{5}{x^2}dx = -5\left[\frac{1}{x}\right]_{20}^{\infty}$$

$$= \boxed{\tfrac{1}{4}}$$

(c) What is the probability that of 6 such devices at least 3 will function for at least 15 years? What assumptions did you make?

Answer:

The solution to this problem requires the assumption that each device is not reliant upon any other, resulting in independence for each. This becomes the sum of probabilities of Binomial r. v., where $n = 6$ and the probability p of any of the individual devices surviving 15 years is given by

$$p = P\{X > 15\} = \int_{15}^{\infty} \frac{5}{x^2} dx = -5 \left[\frac{1}{x}\right]_{15}^{\infty} = \frac{1}{3}$$

Then, to simplify, the calculation can be modified to

$$\sum_{i=3}^{6} P(X = i) = 1 - \sum_{j=0}^{2} P(X = j)$$

$$P(X = 0) = \binom{6}{0} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^6 = \frac{2^6}{3^6}$$

$$P(X = 1) = \binom{6}{1} \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^5 = \frac{6 \times 2^5}{3^6}$$

$$P(X = 2) = \binom{6}{2} \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^4 = \frac{15 \times 2^4}{3^6}$$

$$P(3 \le X \le 6) = 1 - \left(\frac{2^6 + 6 \times 2^5 + 15 \times 2^4}{3^6}\right)$$

$$\approx \boxed{0.3196}$$